

Algebra II with Trigonometry

Chapter 6.2

Olympiad Solutions

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[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

Solutions

1. Without using a calculator, evaluate $\cos(225^\circ) + \sin(495^\circ)$. Quickly justify your reasoning and sign choices.

Solution:

First, reduce the angles to the standard interval 0° to 360° :

- 225° is already between 0° and 360° .
- $495^\circ = 495^\circ - 360^\circ = 135^\circ$.

So:

$$\cos(225^\circ) + \sin(495^\circ) = \cos(225^\circ) + \sin(135^\circ)$$

Now:

- $225^\circ = 180^\circ + 45^\circ$, third quadrant: cosine is negative and has value

$$\cos(225^\circ) = -\frac{\sqrt{2}}{2}.$$

- $135^\circ = 180^\circ - 45^\circ$, second quadrant: sine is positive,

$$\sin(135^\circ) = \sin(45^\circ) = \frac{\sqrt{2}}{2}.$$

So,

$$\cos(225^\circ) + \sin(135^\circ) = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = 0$$

Final Answer: $\boxed{0}$

2. If $\tan t = -\frac{3}{4}$ and t is in Quadrant II, find the exact value of $\sin t + \cos t$.

Solution:

Quadrant II: sine positive, cosine negative.

$$\text{Let } \tan t = \frac{\sin t}{\cos t} = -\frac{3}{4}.$$

Let $\sin t = 3k$, $\cos t = -4k$ (since cosine is negative in II).

$$\sin^2 t + \cos^2 t = 1 \implies (3k)^2 + (-4k)^2 = 1 \implies 9k^2 + 16k^2 = 1 \implies 25k^2 = 1$$

Since sine is positive in Quadrant II, $k > 0$:

$$\text{So } \sin t = 3 \cdot \frac{1}{5} = \frac{3}{5}, \cos t = -4 \cdot \frac{1}{5} = -\frac{4}{5}.$$

$$\sin t + \cos t = \frac{3}{5} - \frac{4}{5} = -\frac{1}{5}$$

Final Answer: $\boxed{-\frac{1}{5}}$

3. Find all t in $0 < t < 2\pi$ such that $\sec t < 0$ and $\cot t > 0$. Express your answer in interval form.

Solution:

$\sec t < 0 \implies \cos t < 0$: this happens in Quadrants II and III.

$\cot t > 0 \implies \tan t > 0$: tangent is positive in Quadrants I and III.

So both occur **only in Quadrant III**.

In Quadrant III,

$$\pi < t < \frac{3\pi}{2}$$

Final Answer: $\boxed{(\pi, \frac{3\pi}{2})}$

4. Let $y = 7 \sin x + 24 \cos x$. What is the maximum value of y as x varies over all real numbers? (No calculus.)

Solution:

This is of the form $a \sin x + b \cos x$, whose maximum is $\sqrt{a^2 + b^2}$.

$$y_{\max} = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25$$

Final Answer: $\boxed{25}$

(Quick justification: The maximum occurs when $\sin x$ and $\cos x$ are in the same proportion as a and b .)

5. Suppose $\sin t + \cos t = 1$. Determine the value of $\sin^4 t + \cos^4 t$.

Solution:

Let $S = \sin t + \cos t = 1$.

Now,

$$\sin^4 t + \cos^4 t = (\sin^2 t + \cos^2 t)^2 - 2 \sin^2 t \cos^2 t$$

But $\sin^2 t + \cos^2 t = 1$.

Let's find $\sin^2 t \cos^2 t$:

First,

$$(\sin t + \cos t)^2 = \sin^2 t + 2 \sin t \cos t + \cos^2 t = 1^2 = 1$$

But $\sin^2 t + \cos^2 t = 1$, so

$$1 + 2 \sin t \cos t = 1 \implies \sin t \cos t = 0$$

Therefore,

$$\sin^4 t + \cos^4 t = 1^2 - 2 \cdot 0 = 1$$

Final Answer: 1

6. Evaluate $\tan(13\pi) + \cot(17\pi)$ mentally, using periodicity and symmetry.

Solution:

Use periodicity:

$$\tan(\theta + n\pi) = \tan \theta$$

And $\tan(\pi) = 0$, $\tan(0) = 0$.

- $13\pi = 12\pi + \pi$, so $\tan(13\pi) = \tan(\pi) = 0$.
- $17\pi = 16\pi + \pi$, so $\cot(17\pi) = \cot(\pi) = 0$.

So overall:

$$\tan(13\pi) + \cot(17\pi) = 0 + 0 = 0$$

Final Answer: 0